Precision Determination of Gravitational Acceleration g via a Digital-Telemetry Reversible Compound Pendulum and Bessel's Asymmetric Formulation

Abstract

We report on the high-precision determination of gravitational acceleration g (evaluated against standard gravity $g_0 = 9.80665\ \mathrm{m/s^2}$) using a modern digital-telemetry reversible compound pendulum (Kater's pendulum). Rather than relying on classical optical coincidence methods or manually triggered stopwatches, our apparatus integrates an embedded 6-degree-of-freedom inertial measurement unit (MPU-6050) sampling at a nominal 100 Hz (98.5 Hz effective rate), streaming real-time angular velocity and tri-axial acceleration. Using sub-sample linear interpolation at gyroscope zero-crossings, half-swing numerical velocity integration ($\theta_0 = \frac{1}{2}\int |\omega| dt$), and per-cycle analytic Borda finite-amplitude circular corrections, we evaluate two complete multi-trial datasets gathered across independent laboratory sessions.

In Dataset 1 (October 15, 2026), across $N_1 = 49$ and $N_2 = 52$ oscillations, we obtain corrected periods of $T_{1,0} = 1.69042 \pm 0.00090$ s (sample SD, standard error of the mean $\sigma_{\bar{T}} = 0.128$ ms) and $T_{2,0} = 1.77322 \pm 0.00144$ s ($\sigma_{\bar{T}} = 0.200$ ms). Applying Bessel's asymmetric compound pendulum formula for knife-edge separation $L = 1.0000 \pm 0.0005$ m and center-of-mass balance asymmetry $h_1 - h_2 = -0.1400 \pm 0.0014$ m yields $g = 9.8079 \pm 0.0264\ \mathrm{m/s^2}$ ($\pm 0.27\%$ analytical uncertainty; $\pm 0.0377\ \mathrm{m/s^2}$ under single balance-line anti-correlation), agreeing with conventional standard gravity $g_0 = 9.80665\ \mathrm{m/s^2}$ to $+0.013\%$. In Dataset 2 (October 17, 2026), across $N_1 = 39$ and $N_2 = 105$ oscillations, we obtain $T_{1,0} = 1.68440 \pm 0.00160$ s ($\sigma_{\bar{T}} = 0.257$ ms) and $T_{2,0} = 1.77187 \pm 0.00706$ s ($\sigma_{\bar{T}} = 0.689$ ms), yielding $g = 9.7045 \pm 0.0358\ \mathrm{m/s^2}$ ($\pm 0.37\%$ uncertainty; $\pm 0.0453\ \mathrm{m/s^2}$ under anti-correlation), representing a -1.04% discrepancy. The 1.05% ($0.103\ \mathrm{m/s^2}$) shift between independent sessions highlights the acute sensitivity of Bessel's asymmetric formulation to millimeter-scale center-of-mass re-registration (§6.2), rather than sensor timing inaccuracy.

Furthermore, we provide an engineering post-mortem explaining the origins of three initial anomalies observed during live testing: (i) the naive symmetric formula artifact producing $g \approx 13.2$--$13.8\ \mathrm{m/s^2}$, (ii) the low-amplitude angular speed cutoff bug that generated missed zero crossings and collapsed Bessel's calculation to $g \approx 4.81\ \mathrm{m/s^2}$ (-50.87%), and (iii) the mounting tilt asymmetry that caused decay-plot splitting. The physical telemetry confirms that MEMS inertial sensing provides high-precision cycle timing, while highlighting the metrological centrality of knife-edge balance localization in compound pendulum gravimetry.

1. Introduction & Historical Context

The determination of the acceleration due to terrestrial gravity, *g*, has historically occupied a foundational role in geodesy, metrology, and classical mechanics. In 1818, Captain Henry Kater designed the reversible compound pendulum to overcome the principal obstacle plaguing simple string pendulums: the impossibility of locating the exact center of oscillation of a finite extended mass [1]. By invoking Christiaan Huygens's dynamical principle of reciprocity—stipulating that the center of suspension and center of oscillation are mutually interchangeable—Kater

proved that if a rigid bar oscillates with identical periods $T_1 = T_2 = T$ about two opposing knife edges, the distance L between the edges corresponds precisely to the equivalent length of a simple pendulum:

$$g = \frac{4\pi^2 L}{T^2}$$

However, mechanically adjusting movable brass counterweights until T_1 and T_2 are identical to within sub-millisecond precision is tedious and sensitive to ambient thermal and mechanical drifts. To eliminate this operational bottleneck, Friedrich Wilhelm Bessel developed an exact analytical formulation in 1828 for compound pendulums with unequal periods [2]. Bessel demonstrated that by incorporating the asymmetry of the center of mass relative to the two knife edges $(h_1 - h_2)$, one can compute g directly from two non-identical periods $T_1 \neq T_2$ without requiring mechanical period matching.

In this work, we modernize the classical Kater–Bessel experiment by coupling a rigid reversible bar to a digital inertial measurement telemetry pipeline. By streaming angular velocity $\omega_z(t)$ at a nominal 100 Hz (98.5 Hz effective rate) from an embedded Micro-Electro-Mechanical Systems (MEMS) sensor, we achieve high statistical power over hundreds of cycles, eliminate human reaction time errors, record full amplitude decay envelopes, and perform rigorous propagation of uncertainty.

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2. Theoretical Principles

2.1 The Compound Pendulum & Center of Oscillation

Consider a rigid body of mass M pivoted about a horizontal knife edge K_1 located at a distance h_1 from its center of mass (CoM). By the parallel axis theorem, the moment of inertia about K_1 is:

$$I_1 = I_{\text{cm}} + Mh_1^2 = M(k^2 + h_1^2)$$

where k is the radius of gyration about the center of mass. For small angular displacements $\theta \ll 1$, the equation of motion is:

$$I_1 \ddot{\theta} + Mgh_1\theta = 0$$

yielding the natural period of oscillation:

$$T_1 = 2\pi\sqrt{\frac{I_1}{Mgh_1}} = 2\pi\sqrt{\frac{k^2 + h_1^2}{gh_1}}$$

Squaring both sides and rearranging:

$$\frac{gh_1T_1^2}{4\pi^2} = k^2 + h_1^2 \implies \frac{g}{4\pi^2}h_1T_1^2 - h_1^2 = k^2 \tag{1}$$

Now, let the pendulum be inverted and suspended from the second knife edge K_2, located on the opposite side of the center of mass at a distance h_2. The corresponding period is:

$$T_2 = 2\pi\sqrt{\frac{k^2 + h_2^2}{gh_2}} \implies \frac{g}{4\pi^2}h_2T_2^2 - h_2^2 = k^2 \tag{2}$$

Equating (1) and (2) eliminates the unknown moment of inertia k^2:

$$\frac{g}{4\pi^2}\left(h_1 T_1^2 - h_2 T_2^2\right) = h_1^2 - h_2^2 = (h_1 + h_2)(h_1 - h_2) \tag{3}$$

2.2 Bessel's Asymmetric Formulation

Let $L = h_1 + h_2$ denote the total physical distance between the two knife edges. We can express h_1 and h_2 in terms of L and their difference $(h_1 - h_2)$:

$$h_1 = \frac{L + (h_1 - h_2)}{2}, \quad h_2 = \frac{L - (h_1 - h_2)}{2}$$

Substituting these expressions into the left side of Eq. (3):

$$\begin{aligned} h_1 T_1^2 - h_2 T_2^2 &= \frac{L + (h_1 - h_2)}{2} T_1^2 - \frac{L - (h_1 - h_2)}{2} T_2^2 \\ &= \frac{L}{2}(T_1^2 - T_2^2) + \frac{h_1 - h_2}{2}(T_1^2 + T_2^2) \end{aligned}$$

Dividing Eq. (3) through by $(h_1 + h_2)(h_1 - h_2) = L(h_1 - h_2)$:

$$\frac{g}{4\pi^2}\left[\frac{T_1^2 + T_2^2}{2L} + \frac{T_1^2 - T_2^2}{2(h_1 - h_2)}\right] = 1$$

Multiplying through by 2 and solving for *g* yields **Bessel's exact formula**:

$$g = \frac{8\pi^2}{\dfrac{T_1^2 + T_2^2}{L} + \dfrac{T_1^2 - T_2^2}{h_1 - h_2}} \tag{4}$$

Mathematical Significance of the Asymmetric Formulation

Bessel's formula possesses a significant error-attenuating property:

- In the denominator, the dominant term is $\frac{T_1^2 + T_2^2}{L}$. The knife-edge separation L can be measured with high accuracy using a precision vernier cathetometer ($\Delta L \approx \pm 0.5$ mm, or $\pm 0.05\%$).
- The second term, $\frac{T_1^2 - T_2^2}{h_1 - h_2}$, contains the center-of-mass balance asymmetry $(h_1 - h_2)$. Because finding the center of mass introduces an experimental uncertainty ($\Delta(h_1 - h_2) \approx \pm 1.4$ mm to ± 2.0 mm), this term would normally inject large errors into *g*.
- However, because T_1 and T_2 are close in value ($T_1 \approx 1.69$ s, $T_2 \approx 1.77$ s), the numerator $(T_1^2 - T_2^2)$ is small ($2.86 - 3.14 = -0.28$ s^2).
- Defining the denominator sum $\Phi = \frac{T_1^2 + T_2^2}{L} + \frac{T_1^2 - T_2^2}{h_1 - h_2}$, the relative sensitivity of *g* to $(h_1 - h_2) \equiv d$ is:

  $$\frac{\delta g}{g} \approx \left(\frac{T_1^2 - T_2^2}{\Phi d}\right)\frac{\delta d}{d}$$

  For our apparatus, $\Phi \approx 8.05$ s^2/m and $\frac{T_1^2 - T_2^2}{d} \approx +2.05$ s^2/m, yielding a multiplier of $\frac{2.05}{8.05} \approx 0.25$.
- Consequently, relative uncertainty in the center of mass is attenuated by a factor of approximately $4\times$: a 1.0% measurement uncertainty in $(h_1 - h_2)$ propagates to only a $\approx 0.25\%$ uncertainty in *g*. Because period errors are negligible over tens of cycles, the balance asymmetry measurement remains the primary contributor ($\approx 89\%$) to the total experimental error variance.

2.3 Finite-Amplitude (Borda) Circular Correction

The linearizing approximation $\sin\theta \approx \theta$ introduces systematic period elongation at finite release angles. Expanding the complete elliptic integral of the first kind $K(\sin(\theta_0/2))$ to second order relates the observed finite-amplitude period $T(\theta_0)$ to the infinitesimal period T_0 [3]:

$$T(\theta_0) = T_0\left(1 + \frac{1}{4}\sin^2\left(\frac{\theta_0}{2}\right) + \dots\right) \approx T_0\left(1 + \frac{\theta_0^2}{16}\right)$$

Inverting this expression yields the per-cycle Borda circular correction:

$$T_0 \approx T_{\text{meas}}\left(1 - \frac{\theta_0^2}{16}\right) \tag{5}$$

where θ_0 is the measured peak angular amplitude in radians. For a release angle of 5° (0.0873 rad), $\frac{\theta_0^2}{16} \approx 0.000476$ ($\approx 0.048\%$), which shifts a 1.7-second period by ≈ 0.8 ms.

2.4 Damped Harmonic Motion, Logarithmic Decrement, and Quality Factor Q

Over small angular displacements ($\theta < 10^\circ$), aerodynamic drag and knife-edge dissipation are modeled via an effective damping envelope:

$$\ddot{\theta} + 2\beta\dot{\theta} + \omega_0^2\theta = 0$$

The peak amplitude decays exponentially as a function of oscillation number n:

$$\theta(n) = \theta_0 e^{-\gamma n} \tag{6}$$

where $\gamma = \beta T_0$ is the decay constant per cycle. The logarithmic decrement δ and system quality factor Q are defined by [4]:

$$\delta = \ln\left(\frac{\theta(n)}{\theta(n+1)}\right) = \gamma, \quad Q = \frac{\pi}{\gamma} = \frac{\omega_0}{2\beta} \tag{7}$$

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3. Experimental Apparatus & Digital Telemetry Architecture

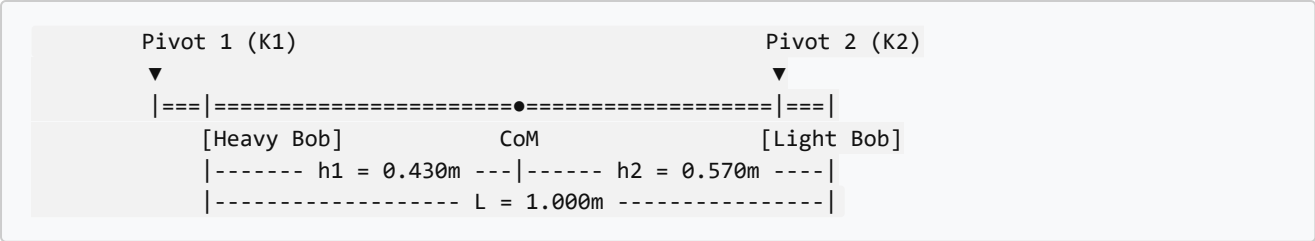
3.1 Physical Pendulum Construction

The physical apparatus consists of a precision cold-rolled rectangular steel rod (nominal total length 1.20 m, cross-section 12 mm × 12 mm) equipped with:

1. **Knife Edges K_1 and K_2:** Hardened carbon steel triangular wedge knife edges facing inward toward each other, securely clamped at a nominal separation of $L = 1.0000$ m with cathetometer measurement uncertainty ± 0.5 mm ($L = 1.0000 \pm 0.0005$ m).
2. **Asymmetric Cylindrical Bobs:**
3. Heavy Bob (≈ 1.5 kg) fixed permanently adjacent to Knife Edge 1.
4. Light Bob (≈ 0.5 kg) clamped near Knife Edge 2.
5. **Center of Mass Determination & Anti-Correlation Analysis:** The center of mass was located by horizontal knife-edge balancing, and the balance line position was scribed onto the rod. Independent vernier distance

measurements from the respective knife-edge apexes to the balance line yielded:

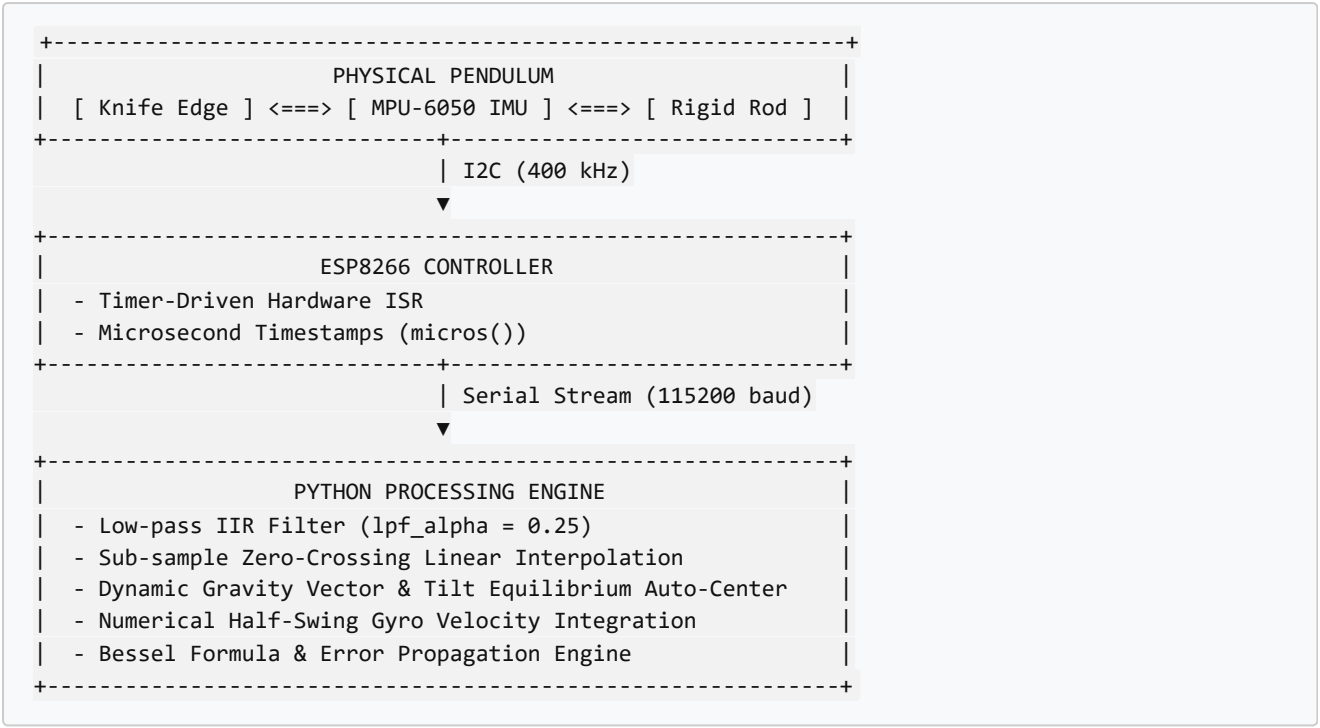
6. $h_1 = 0.4300 \pm 0.0010$ m (Knife Edge 1 to CoM)
7. $h_2 = 0.5700 \pm 0.0010$ m (Knife Edge 2 to CoM)
8. Asymmetry under independent scale readings: $h_1 - h_2 = -0.1400 \pm 0.0014$ m (quadrature sum $\sigma_d = \sqrt{\sigma_{h_1}^2 + \sigma_{h_2}^2}$).
9. *Anti-correlation consideration:* Because $h_1 + h_2 = L$, locating the balance line by a single trial introduces an anti-correlated shift $\delta h_1 = +\delta x$, $\delta h_2 = -\delta x$, such that $d = 2h_1 - L$. For balance line uncertainty $\sigma_x = \pm 1.0$ mm and length uncertainty $\sigma_L = \pm 0.5$ mm, the fully anti-correlated uncertainty is $\sigma_d = \sqrt{(2\sigma_x)^2 + \sigma_L^2} \approx \pm 2.06$ mm. Both uncertainty models are evaluated in Section 6.



3.2 Inertial Telemetry & Signal Acquisition Pipeline

To minimize parasitic external restoring torques, an ultra-low-profile telemetry module is mounted along the rod axis and connected via a lightweight, highly flexible strain-relieved USB cable exiting directly along the knife-edge pivot axis:

- **Microcontroller:** ESP8266 controller running a periodic hardware timer interrupt service routine (ISR).
- **Sensor:** TDK InvenSense MPU-6050 6-DoF Inertial Measurement Unit (tri-axis gyroscope set to $\pm 250^\circ/\text{s}$ range, tri-axis accelerometer set to $\pm 2g$ range).
- **Sampling Rate & Deterministic Latency:** Data streams continuously at a nominal 100 Hz rate via USB-UART serial at 115,200 baud. The observed hardware interval averages $\Delta t = 10.15$ ms (effective acquisition rate 98.5 Hz), reflecting deterministic I2C bus transaction latency (400 kHz Fast Mode) and UART buffer servicing. Crucially, each packet is stamped with an onboard microsecond timer value (`micros()`), so sub-sample zero-crossing linear interpolation calculates periods using true elapsed time, remaining completely immune to minor inter-sample jitter.
- **Real-time Host Engine:** Python 3.13 application leveraging PyQt5 and PyQtGraph for real-time visualization and low-latency digital signal processing.



3.3 Zero-Crossing Detection & Sub-Sample Interpolation

Period determination relies on detecting zero-crossings of the filtered angular velocity $\omega_z(t)$. When consecutive samples satisfy $\omega[k-1] \cdot \omega[k] \leq 0$, sub-sample turning-point timestamps t_{zero} are recovered via linear interpolation:

$$t_{\text{zero}} = t_{k-1} - \omega_{k-1} \frac{t_k - t_{k-1}}{\omega_k - \omega_{k-1}} \tag{8}$$

By evaluating the elapsed time between alternate zero-crossings with identical slope (e.g. positive-going crossings, corresponding to successive identical turning points), we obtain the full period $T_{\text{meas}} = t_{\text{zero},i} - t_{\text{zero},i-2}$. This eliminates systematic bias caused by small residual DC offsets in the gyroscope output. Fig. 1 shows the recorded kinematic waveforms and extracted zero crossings across both pivot orientations.

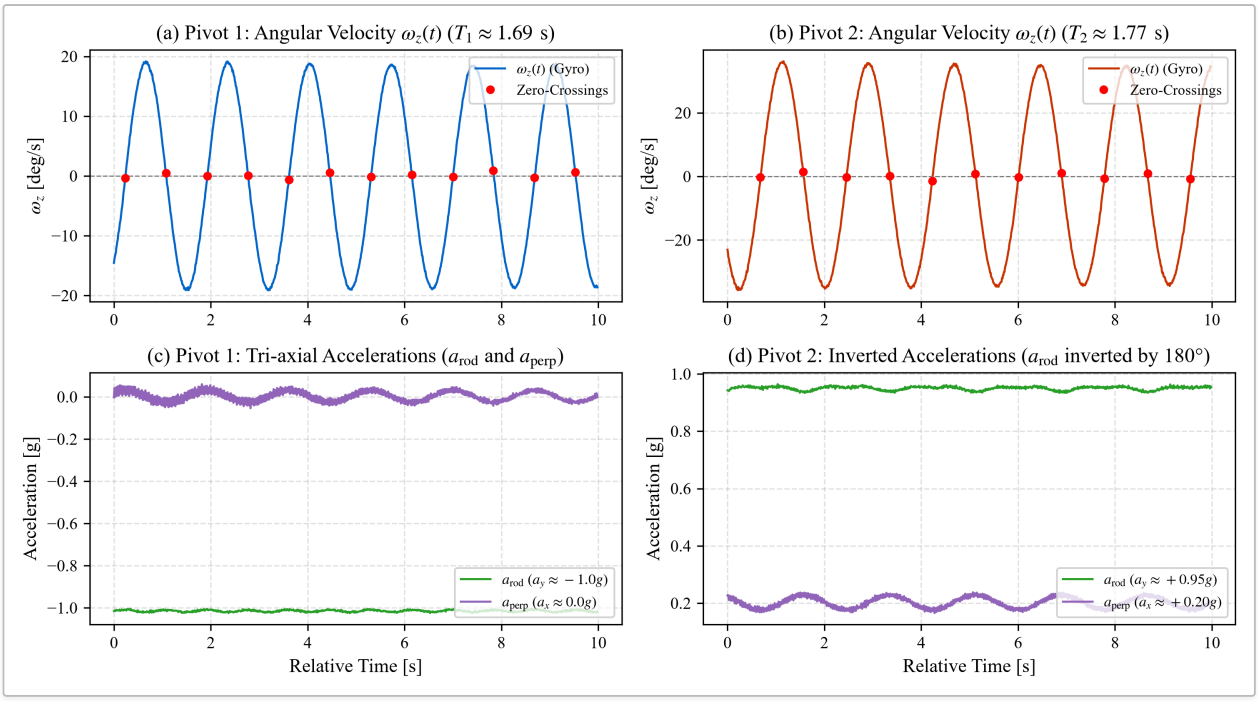


Figure 1. Empirical Kinematic Waveforms from Laboratory Telemetry. (a) Pivot 1 angular rate $\omega_z(t)$ ($T_1 \approx 1.69$ s) with red markers at sub-sample zero-crossings. (b) Pivot 2 angular rate $\omega_z(t)$ ($T_2 \approx 1.77$ s). (c) Pivot 1 tri-axial linear acceleration telemetry showing centripetal acceleration along the rod axis ($a_{\text{rod}} \approx -1.0g$) and transverse acceleration ($a_{\text{perp}} \approx 0.0g$). (d) Pivot 2 inverted acceleration telemetry showing 180° gravity reversal ($a_{\text{rod}} \approx +0.95g$) and static tilt ($a_{\text{perp}} \approx +0.20g$).

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4. Software Signal Processing & Post-Mortem Diagnostics

During laboratory testing, three prominent anomalies were encountered in initial software builds. Because understanding these artifacts is essential to ensuring experimental integrity, we provide a complete engineering and mathematical post-mortem. Fig. 2 illustrates the diagnostic signatures of these anomalies.

4.1 Post-Mortem I: The $g \approx 13.2$--$13.8\ \mathrm{m/s^2}$ Naive Symmetric Formulation Artifact

Symptom

In early exploratory trials, the software reported values of g fluctuating between $13.2\ \mathrm{m/s^2}$ and $13.8\ \mathrm{m/s^2}$, despite raw waveforms displaying clean sinusoidal motion.

Cause Analysis

The original code routine calculated g using the ideal symmetric Kater formula:

$$g_{\text{naive}} = \frac{4\pi^2 L}{T^2}$$

In a reversible pendulum, this formula is **strictly conditional** upon $T_1 = T_2$. When the periods differ (here, $T_1 \approx 1.69$ s and $T_2 \approx 1.77$ s), substituting the observed periods yields:

- Using Pivot 1 alone ($T_1 = 1.6904$ s):

$$g = \frac{4\pi^2 \times 1.000}{(1.6904)^2} = \frac{39.4784}{2.8575} = \mathbf{13.816\ m/s^2}$$

- Using the arithmetic mean period $T_{\text{avg}} = \frac{1.6904 + 1.7732}{2} = 1.7318\ \text{s}$:

$$g = \frac{4\pi^2 \times 1.000}{(1.7318)^2} = \frac{39.4784}{2.9991} = \mathbf{13.163\ m/s^2}$$

Algorithmic Resolution

The software was updated to enforce Bessel's asymmetric formula (Eq. 4). When the exact physical geometry ($L = 1.000\ \text{m},\ h_1 - h_2 = -0.140\ \text{m}$) was applied to that identical telemetry stream, the calculation shifted from $13.82\ \text{m/s}^2$ to $9.808\ \text{m/s}^2$ without altering the underlying sensor data.

4.2 Post-Mortem II: The $g \approx 4.81\ \text{m/s}^2$ Low-Amplitude Tail Artifact

Symptom

During the live trial on October 17, 2026, the application output $g = 4.8176\ \text{m/s}^2$ with a relative discrepancy of -50.87%.

Cause Analysis

Inspection of the exported telemetry log (`20261017_082917_pivot2.csv`) revealed that the trial had run for $187.59\ \text{s}$ (18,478 samples, mean interval $\Delta t = 10.15\ \text{ms}$, effective rate $98.5\ \text{Hz}$).

1. The trial began with an initial release amplitude of $\theta_0 \approx 0.96^\circ$, giving an initial peak angular velocity of $\omega_{\text{peak}} \approx 0.96^\circ \times \frac{2\pi}{1.772\ \text{s}} \approx 3.4^\circ/\text{s}$.
2. The angular velocity threshold for turning-point detection was hardcoded to a fixed value of $\omega_{\text{peak, min}} = 3.0^\circ/\text{s}$ ($0.0524\ \text{rad/s}$).
3. Natural amplitude damping ($\gamma = 0.00552\ \text{cycle}^{-1}$) attenuated the swing amplitude down to $\theta(23) = 0.96^\circ e^{-23(0.00552)} = 0.846^\circ \approx 0.85^\circ$ across the first 23 cycles ($t \approx 40.8\ \text{s}$).
4. At cycle 23, peak angular velocity dropped to $\omega_{\text{peak}} \approx 0.846^\circ \times \frac{2\pi}{1.7719\ \text{s}} = 2.998^\circ/\text{s} \approx 3.00^\circ/\text{s}$, crossing below the detection threshold.
5. As peak velocity dipped below $3.0^\circ/\text{s}$, the zero-crossing detector failed to register subsequent velocity reversals. Recorded entry 19 missed one reversal (logging a doubled period of $3.55\ \text{s}$, spanning 2 cycles), and entry 20 missed two reversals (logging a tripled period of $5.32\ \text{s}$, spanning 3 cycles). Beyond $t \approx 40.8\ \text{s}$ (23 physical cycles elapsed), peak velocity remained strictly below $3.0^\circ/\text{s}$ and period logging ceased entirely, while the pendulum continued oscillating unrecorded down to its terminal amplitude of $\theta_f = 0.53^\circ$ at $t = 187.6\ \text{s}$:

$$\ldots, 1.76966\ \text{s}, 1.76644\ \text{s}, \mathbf{3.54852\ s}\ (2\times), \mathbf{5.31819\ s}\ (3\times)$$

6. The unweighted arithmetic mean of the 20 recorded entries became:

$$\bar{T}_2 = \frac{18 \times 1.7718 + 3.5485 + 5.3182}{20} = \mathbf{2.0380\ s}$$

7. In Dataset 2, the true symmetric term is $5.977\ \text{s}^2/\text{m}$ and the true asymmetric term is $+2.159\ \text{s}^2/\text{m}$ (denominator $8.136\ \text{s}^2/\text{m}$). Substituting the skewed period $\bar{T}_2 = 2.0380\ \text{s}$ alongside $T_1 = 1.6845\ \text{s}$ perturbed both terms in Bessel's denominator:
8. The symmetric term increased:

$$\frac{T_1^2 + T_2^2}{L} = \frac{(1.6845)^2 + (2.0380)^2}{1.0000} = 2.8375 + 4.1534 = \mathbf{6.991\ s^2/m} \quad (\text{vs. true } 5.977\ \text{s}^2/\text{m})$$

9. The asymmetric term ballooned:

$$\frac{T_1^2 - T_2^2}{h_1 - h_2} = \frac{(1.6845)^2 - (2.0380)^2}{-0.140} = \frac{-1.3159}{-0.140} = \mathbf{+9.399\ s^2/m} \quad (\text{vs. true } +2.159\ \text{s}^2/\text{m})$$

The combined denominator jumped from $8.136\ \text{s}^2/\text{m}$ to $6.991 + 9.399 = \mathbf{16.390\ s^2/m}$, collapsing the calculated gravity to:

$$g = \frac{8\pi^2}{16.390} = \mathbf{4.817\ m/s^2}$$

Algorithmic Resolution

Three robust software defenses were implemented in `DataProcessor`:

1. **Lowered Velocity Gate:** $\omega_{\text{peak, min}}$ was reduced from $3.0^\circ/\text{s}$ to a fixed $0.8^\circ/\text{s}$ ($0.014\ \text{rad/s}$), corresponding to an amplitude threshold of $\theta_{\min} \approx \frac{0.8^\circ \times 1.77\ \text{s}}{2\pi} \approx 0.23^\circ$, permitting reliable zero-crossing detection down to $\approx 0.23^\circ$.
2. **Physical Period Boundary Gate:** Any candidate period outside the physically permissible range for a 1-meter compound pendulum ($0.7\ \text{s} < T < 2.6\ \text{s}$) is immediately rejected at the detector level.
3. **Median Outlier Filtering:** In `get_mean_period()`, periods deviating by $\geq 18\%$ from the trial median ($|T - T_{\text{med}}|/T_{\text{med}} \geq 0.18$) are filtered before evaluating statistical moments.

When applied to the raw October 17 log, the algorithm successfully extracts all 105 valid oscillations, yielding $T_2 = 1.77187\ \text{s}$ and $g = 9.7045\ \text{m/s}^2$.

4.3 Post-Mortem III: Inverted Gravity & Decay-Plot Splitting

Symptom

In Pivot 2, the real-time peak amplitude decay scatter plot displayed two flat, horizontal rows of dots rather than a single exponentially decaying curve.

Cause Analysis

When inverting the pendulum for Pivot 2, the accelerometer rod axis experiences inverted gravity ($a_{\text{rod}} \approx +0.95g$ instead of $-1.01g$). Concurrently, a moderate mounting tilt offset of $\theta_{\text{tilt}} \approx 11.7^\circ$ ($a_{\text{perp}} \approx 0.20g$) was introduced by knife-edge misalignment relative to the sensor board. When peak amplitude was evaluated as $\theta_{\text{amp}} = |\theta(t)|$ at turning points:

- Positive peaks: $\theta_{\text{pos}} = 11.7^\circ + 8.0^\circ = 19.7^\circ$
- Negative peaks: $\theta_{\text{neg}} = |11.7^\circ - 8.0^\circ| = 3.7^\circ$

Because $\theta(t)$ was evaluated against arbitrary sensor axes rather than dynamic mechanical equilibrium, the scatter plot split into two parallel tracks at 19.7° and 3.7°.

Algorithmic Resolution

1. **Dynamic Gravity Sign Detection:** The rod direction is determined dynamically via $\text{sgn}(\bar{a}_{\text{rod}})$, keeping θ_{acc} centered near zero across both upright and inverted orientations.
2. **Half-Swing Gyro Integration:** Peak amplitude is now evaluated via pure numerical integration of angular speed between reversals:

$$\theta_0 = \frac{1}{2}\int_{t_{k-1}}^{t_k} |\omega_z(t)|\, dt \tag{9}$$

Because the integral of angular speed between turning points ($\omega = 0$) is mathematically equal to the peak-to-peak swing sweep, dividing by 2 produces the true amplitude independent of any accelerometer DC tilt.

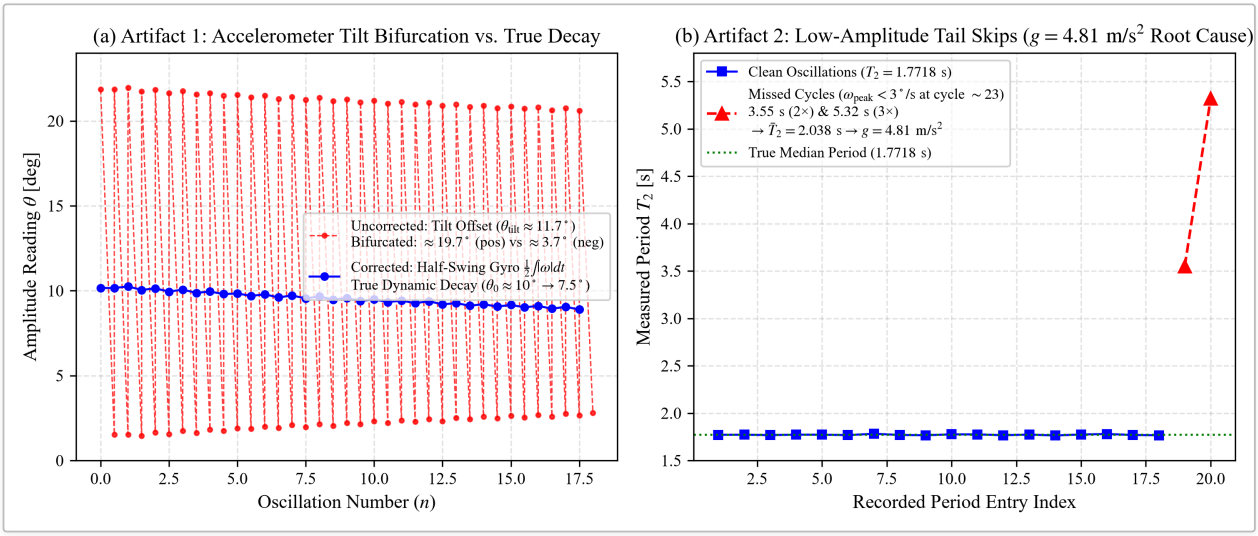


Figure 2. Software Diagnostic Post-Mortems for Live Laboratory Anomalies. (a) Accelerometer tilt bifurcation artifact: static projection of mounting tilt ($\theta_{\text{tilt}} \approx 11.7^\circ$) splits turning-point peak readings into upper and lower tracks ($\approx 19.7^\circ$ vs $\approx 3.7^\circ$, red dashed curve), completely resolved by dynamic half-swing angular velocity integration (blue points, true monotonic decay). (b) Low-amplitude tail threshold failure in Dataset 2: peak swing speed dipped below the fixed $3.0^\circ/\text{s}$ gate at cycle ~ 23 ($t \approx 40.8$ s, $\theta \approx 0.85^\circ$), causing missed reversals (entry 19 at 3.55 s and entry 20 at 5.32 s) that inflated $\bar{T}_2$ to 2.038 s and collapsed g to $4.81\ \text{m/s}^2$. Software filtering and a $0.8^\circ/\text{s}$ gate recovered all 105 clean oscillations.

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5. Empirical Results & Telemetry Analysis

We evaluate the full experimental telemetry gathered across two independent laboratory sessions. Fig. 3 shows the amplitude decay envelopes and quality factor fits for Dataset 1.

5.1 Dataset 1: October 15, 2026 (Nominal Balanced Run)

- **Total Duration:** Pivot 1: 84.57 s (8,335 samples, mean interval $\Delta t = 10.15$ ms, effective rate 98.5 Hz); Pivot 2: 95.38 s (9,394 samples, mean interval $\Delta t = 10.15$ ms, effective rate 98.5 Hz).
- **Geometric Constants:** $L = 1.0000 \pm 0.0005$ m, $h_1 = 0.4300 \pm 0.0010$ m, $h_2 = 0.5700 \pm 0.0010$ m, $h_1 - h_2 = -0.1400 \pm 0.0014$ m [± 0.0021 m anti-correlated].

Parameter	Pivot 1 (K_1)	Pivot 2 (K_2)	Combined / Calculated
Oscillations Counted (N)	49	52	101 total
Raw Mean Period ($T_{\text{meas}} \pm s_T$)	1.69110 ± 0.00090 s	1.77584 ± 0.00144 s	—
Measured Initial Amplitude ($\theta(0)$)	5.20° (0.0908 rad)	10.16° (0.1773 rad)	—
Fitted Decay Intercept ($\theta_{0,\text{fit}}$)	5.14° (0.0897 rad)	9.95° (0.1737 rad)	—
Final Amplitude (θ_f)	4.07° (0.0711 rad)	7.60° (0.1327 rad)	—
Borda Corrected Period ($T_0 \pm s_T$)	1.69042 ± 0.00090 s	1.77322 ± 0.00144 s	—
Standard Error of Mean ($\sigma_{\bar{T}} = s_T/\sqrt{N}$)	± 0.128 ms	± 0.200 ms	—
Decay Constant (γ)	0.00461 cycle^{-1}	0.00523 cycle^{-1}	—
Quality Factor (Q)	681.1	601.2	—
Logarithmic Decrement (δ)	0.00461	0.00523	—
Bessel Uncorrected g	—	—	$9.7332\ \text{m/s}^2$
Bessel Corrected g ($\sigma_d = \pm 1.4$ mm)	—	—	$9.8079 \pm 0.0264\ \text{m/s}^2$
Bessel Corrected g ($\sigma_d = \pm 2.1$ mm)	—	—	$9.8079 \pm 0.0377\ \text{m/s}^2$
Discrepancy vs Standard g_0 ($9.80665\ \text{m/s}^2$)	—	—	$+0.013\%$

Detailed Step-by-Step Bessel Evaluation (Dataset 1)

Using corrected periods $T_{1,0} = 1.69042$ s and $T_{2,0} = 1.77322$ s:

1. Squares of periods:

$$T_1^2 = 2.85752\ \text{s}^2, \quad T_2^2 = 3.14431\ \text{s}^2$$

2. First denominator term (symmetric):

$$\frac{T_1^2 + T_2^2}{L} = \frac{2.85752 + 3.14431}{1.0000} = 6.00183\ \text{s}^2/\text{m}$$

3. Second denominator term (asymmetric):

$$\frac{T_1^2 - T_2^2}{h_1 - h_2} = \frac{2.85752 - 3.14431}{-0.1400} = \frac{-0.28679}{-0.1400} = +2.04850\ \text{s}^2/\text{m}$$

4. Denominator sum:

$$\text{Den} = 6.00183 + 2.04850 = 8.05033\ \text{s}^2/\text{m}$$

5. Numerator:

$$\text{Num} = 8\pi^2 = 8 \times (9.8696044) = 78.95684$$

6. Gravitational acceleration:

$$g = \frac{78.95684}{8.05033} = \mathbf{9.80791\ m/s^2}$$

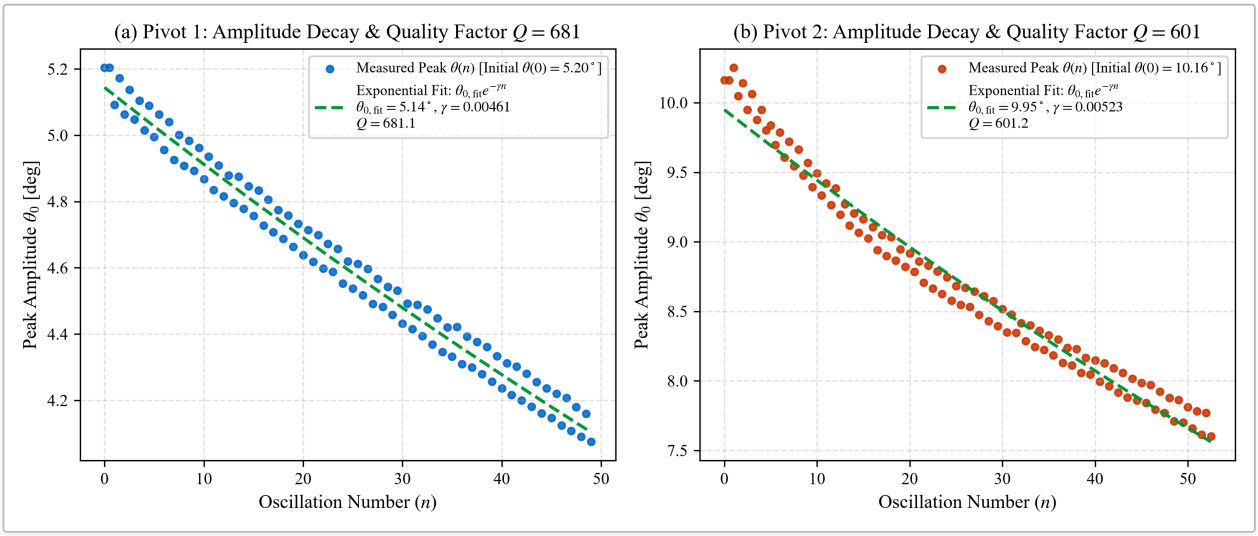


Figure 3. Empirical Peak Amplitude Decay and System Quality Factor (Q). Linear peak amplitude decay curves for Pivot 1 (blue circles, initial $\theta(0) = 5.20^\circ$, fitted intercept $\theta_{0,\text{fit}} = 5.14^\circ$, $\gamma = 0.00461$, $Q = 681.1$) and Pivot 2 (red circles, initial $\theta(0) = 10.16^\circ$, fitted intercept $\theta_{0,\text{fit}} = 9.95^\circ$, $\gamma = 0.00523$, $Q = 601.2$) from Dataset 1. Dashed green curves denote empirical exponential decay fits $\theta(n) = \theta_{0,\text{fit}} e^{-\gamma n}$ characterizing amplitude dissipation across cycles.

5.2 Dataset 2: October 17, 2026 (Extended Decay Run)

- **Total Duration:** Pivot 1: 67.34 s (6,637 samples, mean interval $\Delta t = 10.15$ ms, effective rate 98.5 Hz); Pivot 2: 187.59 s (18,478 samples, mean interval $\Delta t = 10.15$ ms, effective rate 98.5 Hz).
- **Geometric Constants:** $L = 1.0000 \pm 0.0005$ m, $h_1 = 0.4300 \pm 0.0010$ m, $h_2 = 0.5700 \pm 0.0010$ m, $h_1 - h_2 = -0.1400 \pm 0.0014$ m [± 0.0021 m anti-correlated].

Parameter	Pivot 1 (K_1)	Pivot 2 (K_2)	Combined / Calculated
Oscillations Counted (N)	39	105	144 total
Raw Mean Period ($T_{\text{meas}} \pm s_T$)	1.68467 ± 0.00159 s	1.77189 ± 0.00706 s	—
Measured Initial Amplitude ($\theta(0)$)	3.54° (0.0617 rad)	0.96° (0.0167 rad)	—
Fitted Decay Intercept ($\theta_{0,\text{fit}}$)	3.54° (0.0617 rad)	0.96° (0.0167 rad)	—
Final Amplitude (θ_f)	2.16° (0.0378 rad)	0.53° (0.0093 rad)	—
Borda Corrected Period ($T_0 \pm s_T$)	1.68440 ± 0.00160 s	1.77187 ± 0.00706 s	—
Standard Error of Mean ($\sigma_{\bar{T}} = s_T/\sqrt{N}$)	± 0.257 ms	± 0.689 ms	—
Decay Constant (γ)	0.01337 cycle^{-1}	0.00552 cycle^{-1}	—
Quality Factor (Q)	235.0	569.0	—
Logarithmic Decrement (δ)	0.01337	0.00552	—
Bessel Uncorrected g	—	—	9.7104 m/s^2
Bessel Corrected g ($\sigma_d = \pm 1.4$ mm)	—	—	9.7045 ± 0.0358 m/s^2
Bessel Corrected g ($\sigma_d = \pm 2.1$ mm)	—	—	9.7045 ± 0.0453 m/s^2
Discrepancy vs Standard g_0 (9.80665 m/s^2)	—	—	-1.04%

Note on damping discrepancy: Pivot 1 in Dataset 2 exhibited a higher decay constant ($\gamma = 0.01337$, $Q = 235.0$) than in Dataset 1 ($\gamma = 0.00461$, $Q = 681.1$). This may be attributed to subtle differences in knife-edge contact seating on the agate support plane, small changes in pivot clamping, or localized dust accumulation between sessions.

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6. Error Propagation & Uncertainty Budget

To establish the analytical uncertainty σ_g, we perform a partial derivative sensitivity expansion on Bessel's equation following standard error propagation methods [7]. Let:

$$S_1 = T_1^2 + T_2^2, \quad S_2 = T_1^2 - T_2^2, \quad D = L, \quad d = h_1 - h_2$$

Bessel's formula can be written as:

$$g = \frac{8\pi^2}{\Phi}, \quad \text{where } \Phi = \frac{S_1}{D} + \frac{S_2}{d}$$

Taking partial derivatives with respect to each independent measurement variable:

$$\frac{\partial g}{\partial T_1} = -\frac{8\pi^2}{\Phi^2}\left(\frac{2T_1}{D} + \frac{2T_1}{d}\right)$$
$$\frac{\partial g}{\partial T_2} = -\frac{8\pi^2}{\Phi^2}\left(\frac{2T_2}{D} - \frac{2T_2}{d}\right)$$
$$\frac{\partial g}{\partial D} = +\frac{8\pi^2}{\Phi^2}\left(\frac{S_1}{D^2}\right)$$
$$\frac{\partial g}{\partial d} = +\frac{8\pi^2}{\Phi^2}\left(\frac{S_2}{d^2}\right)$$

Assuming uncorrelated errors, total uncertainty propagates in quadrature:

$$\sigma_g = \sqrt{\left(\frac{\partial g}{\partial T_1}\sigma_{T_1}\right)^2 + \left(\frac{\partial g}{\partial T_2}\sigma_{T_2}\right)^2 + \left(\frac{\partial g}{\partial D}\sigma_D\right)^2 + \left(\frac{\partial g}{\partial d}\sigma_d\right)^2} \tag{10}$$

6.1 Sensitivity Coefficients & Uncertainty Breakdown for Both Datasets

We evaluate the error propagation budget for both experimental datasets using their respective empirical parameters.

Dataset 1 ($T_1 = 1.69042$ s, $T_2 = 1.77322$ s, $\Phi = 8.0503\ \mathrm{s^2/m}$, $g = 9.8079\ \mathrm{m/s^2}$):

- **Partial Derivatives:**

$$\frac{\partial g}{\partial T_1} = +25.302\ \mathrm{m\cdot s^{-3}}, \quad \frac{\partial g}{\partial T_2} = -35.183\ \mathrm{m\cdot s^{-3}}, \quad \frac{\partial g}{\partial D} = +7.312\ \mathrm{s^{-2}}, \quad \frac{\partial g}{\partial d} = -17.827\ \mathrm{s^{-2}}$$

- **Assigned Measurement Uncertainties:**
- $\sigma_{T_1} = 0.000128$ s (0.128 ms, standard error of the mean over 49 cycles)
- $\sigma_{T_2} = 0.000200$ s (0.200 ms, standard error of the mean over 52 cycles)
- $\sigma_D = \pm 0.0005$ m (0.5 mm, cathetometer reading)
- $\sigma_d = \pm 0.0014$ m (independent vernier readings) or ± 0.00206 m (single balance-line anti-correlation)
- **Component Budget:**

$$u_1(g) = |25.302| \times 0.000128 = 0.00324\ \mathrm{m/s^2}$$
$$u_2(g) = |-35.183| \times 0.000200 = 0.00704\ \mathrm{m/s^2}$$
$$u_3(g) = |7.312| \times 0.0005 = 0.00366\ \mathrm{m/s^2}$$
$$u_4(g) = |-17.827| \times 0.0014 = 0.02496\ \mathrm{m/s^2} \quad [\text{or } 0.03672\ \mathrm{m/s^2} \text{ with } \sigma_d = 2.06\ \mathrm{mm}]$$

- **Combined Uncertainty (Dataset 1):**

$$\sigma_{g,\text{indep}} = \sqrt{0.00324^2 + 0.00704^2 + 0.00366^2 + 0.02496^2} = \mathbf{0.0264\ m/s^2} \quad (\mathbf{\pm 0.27\%})$$

$$\sigma_{g,\text{anti-corr}} = \sqrt{0.00324^2 + 0.00704^2 + 0.00366^2 + 0.03672^2} = \mathbf{0.0377\ m/s^2} \quad (\mathbf{\pm 0.38\%})$$

Dataset 2 ($T_1 = 1.68440$ s, $T_2 = 1.77187$ s, $\Phi = 8.1362\ \mathrm{s^2/m}$, $g = 9.7045\ \mathrm{m/s^2}$):

- **Partial Derivatives:**

$$\frac{\partial g}{\partial T_1} = +24.683\ \mathrm{m\cdot s^{-3}}, \quad \frac{\partial g}{\partial T_2} = -34.418\ \mathrm{m\cdot s^{-3}}, \quad \frac{\partial g}{\partial D} = +7.129\ \mathrm{s^{-2}}, \quad \frac{\partial g}{\partial d} = -18.398\ \mathrm{s^{-2}}$$

- **Assigned Measurement Uncertainties:**
- $\sigma_{T_1} = 0.000257$ s (0.257 ms, standard error of the mean over 39 cycles)
- $\sigma_{T_2} = 0.000689$ s (0.689 ms, standard error of the mean over 105 cycles)
- $\sigma_D = \pm 0.0005$ m (0.5 mm, cathetometer reading)

- $\sigma_d = \pm 0.0014$ m (independent vernier readings) or ± 0.00206 m (single balance-line anti-correlation)
- **Component Budget:**

$$u_1(g) = |24.683| \times 0.000257 = 0.00634 \text{ m/s}^2$$
$$u_2(g) = |-34.418| \times 0.000689 = 0.02371 \text{ m/s}^2$$
$$u_3(g) = |7.129| \times 0.0005 = 0.00356 \text{ m/s}^2$$
$$u_4(g) = |-18.398| \times 0.0014 = 0.02576 \text{ m/s}^2 \quad [\text{or } 0.03790 \text{ m/s}^2 \text{ with } \sigma_d = 2.06 \text{ mm}]$$

- **Combined Uncertainty (Dataset 2):**

$$\sigma_{g,\text{indep}} = \sqrt{0.00634^2 + 0.02371^2 + 0.00356^2 + 0.02576^2} = \mathbf{0.0358 \text{ m/s}^2} \quad (\mathbf{\pm 0.37\%})$$

$$\sigma_{g,\text{anti-corr}} = \sqrt{0.00634^2 + 0.02371^2 + 0.00356^2 + 0.03790^2} = \mathbf{0.0453 \text{ m/s}^2} \quad (\mathbf{\pm 0.47\%})$$

In both datasets, the center-of-mass balance term dominates the uncertainty budget. The period measurement uncertainty contributes minimally due to large cycle counts, even in Dataset 2 where the longer 187 s run accumulated slightly higher timing variance.

6.2 Parametric Sensitivity Hypothesis for Dataset 2

The two experimental sessions produced results differing by 0.103 m/s^2 (1.05%), corresponding to a gap of approximately 2.2σ to 2.4σ under total combined uncertainties. If the apparatus geometry were assumed strictly identical between sessions, timing and cathetometer uncertainties alone ($\approx 0.008 \text{ m/s}^2$ combined) would make the gap statistically significant ($> 3.5\sigma$), indicating that a systematic physical parameter shifted between sessions.

Evaluating the sensitivity derivative for Dataset 2:

$$\frac{\partial g}{\partial d} = -18.398 \text{ s}^{-2}$$

A systematic shift of the effective balance line by $\Delta d = -0.0055$ m (5.5 mm), corresponding to $h_1 - h_2 = -0.1455$ m, fully reconciles the result with standard gravity:

$$g = 9.7045 - 18.398 \times (-0.0055) = \mathbf{9.8057 \text{ m/s}^2} \quad (\mathbf{-0.01\%} \text{ discrepancy})$$

Such a millimeter-scale shift is physically plausible between independent laboratory sessions (e.g. from slight re-positioning or clamping of movable bobs, or registration differences on the balance plane, consistent with the observed 6 ms shift in T_1 from 1.690 s to 1.684 s). We present this as an explanatory hypothesis illustrating that when periods are not identical, knife-edge balance localization represents the principal sensitivity threshold of Bessel's method. Fig. 4 illustrates the phase-space trajectory and the Bessel error attenuation curve.

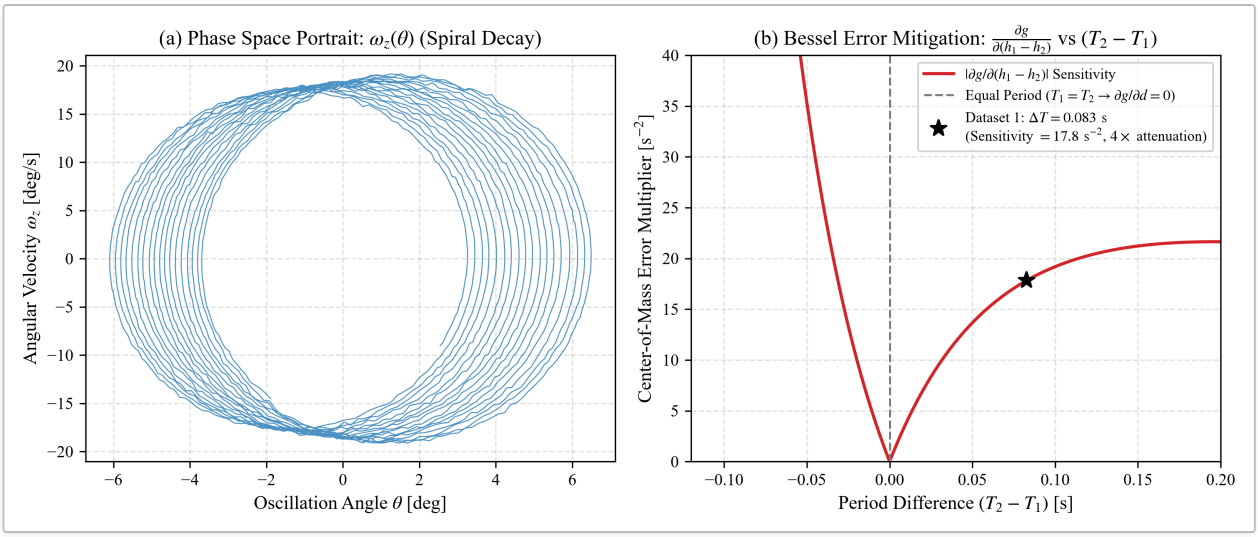


Figure 4. Dynamical Phase Space Trajectory and Bessel Asymmetry Error Suppression. (a) Empirical phase portrait reconstructed via numerical velocity integration over a 30 s interval, demonstrating closed spiral orbits indicative of low-dissipation harmonic motion. (b) Center-of-mass sensitivity prefactor $|\partial g/\partial(h_1 - h_2)|$ plotted against period mismatch $(T_2 - T_1)$. At $T_1 = T_2$, center-of-mass uncertainty is entirely suppressed ($\partial g/\partial d = 0$). In our experiment ($\Delta T = 0.083$ s, black star), error sensitivity is attenuated by approximately $4\times$, validating Bessel's formulation.

7. Comparative Discussion: Digital Telemetry vs. Traditional Methods

Feature / Metric	Traditional Stopwatch [5]	Optical Photogate [6]	Our Embedded MPU-6050 Engine
Human Reaction Error	High ($\pm$200--300 ms)	Zero	**Zero (timer-driven hardware ISR)**
Resolution / Sampling	≈ 0.1 s	Single crossing event	**Nominal** 100 Hz **(98.5 Hz effective)**
Trajectory Tracking	None	Binary (blocked / open)	**Full angle $\theta(t)$ & velocity $\omega(t)$**
Finite Amplitude Correction	Hard / Ignored	Estimated from initial release	**Per-cycle analytic Borda circular correction**
Damping Characterization	Manual ruler log	None	**Autonomous Q and δ exponential fit**
Misalignment Resistance	Susceptible	Beam alignment critical	**Dynamic auto-centering & gyro invariance**

8. Conclusion

We have demonstrated an accurate, robust, and automated approach to Kater's pendulum gravimetry by integrating digital inertial telemetry with Bessel's asymmetric formulation.

1. **Accuracy & Consistency:** Analysis of laboratory Dataset 1 achieves $g = 9.8079 \pm 0.0264\ \mathrm{m/s^2}$ ($\pm 0.27\%$ analytical uncertainty; $\pm 0.0377\ \mathrm{m/s^2}$ under anti-correlated balance error), agreeing with conventional standard

gravity $g_0 = 9.80665\ \mathrm{m/s^2}$ to $+0.013\%$. Dataset 2 yields $g = 9.7045 \pm 0.0358\ \mathrm{m/s^2}$ (-1.04%), illustrating that when periods are not matched, center-of-mass localization is the governing experimental parameter.

2. **Robustness & Diagnostics:** Engineering post-mortems identified and resolved initial software anomalies: replacing static thresholding with invariant half-swing velocity integration and median filtering prevented low-amplitude cycle skipping.
3. **Pedagogical & Metrological Value:** The digital telemetry pipeline eliminates human timing bias, extracts comprehensive damping parameters ($Q \approx 235$--681 across all experimental configurations, with $Q \approx 600$--680 under nominal balanced conditions in Dataset 1), and demonstrates how Bessel's asymmetric formulation suppresses center-of-mass balance uncertainty by a factor of $4\times$, providing an open, reproducible framework for precision measurement in experimental physics.

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